

CSL7590: Deep Learning

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Generative Models

Given a set of points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ from a distribution $p_{\text{data}}(\mathbf{x})$. Learn a distribution $p_{\text{model}}(\mathbf{x})$ from the given samples such that the learned distribution is as close as possible to the ground truth distribution. How to sample new points from the learned distribution $p_{\text{model}}(\mathbf{x})$?



pc:nyu

Attribute	Value
Spectacles	Yes/No
Expression	Sad/Angry/Smiling/..
Beard	Yes/No
Age	1,2,3,...
Viewpoint	{0,1,2,3,...,180}
Nose Style	Long/Medium/Short
Hair Style	Long/Short
...	...

We want a continuous parameter for a controlled sampling.

Generative Process

Given a set of points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ from a distribution $p_{\text{data}}(\mathbf{x})$,
First learn a distribution $p_{\text{model}}(\mathbf{x})$ from the given samples such
that the learned distribution is as close as possible to the ground
truth distribution. How to sample new points from the learned
distribution $p_{\text{model}}(\mathbf{x})$?

Let us call these control parameters as latent parameters

$\mathbf{z} = [z_1 \ z_2 \ \dots \ z_k]^\top$. Then,

$$p(\mathbf{x}) = p(\mathbf{x})$$

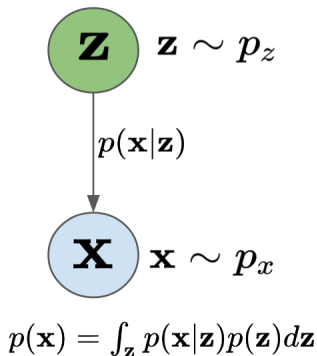
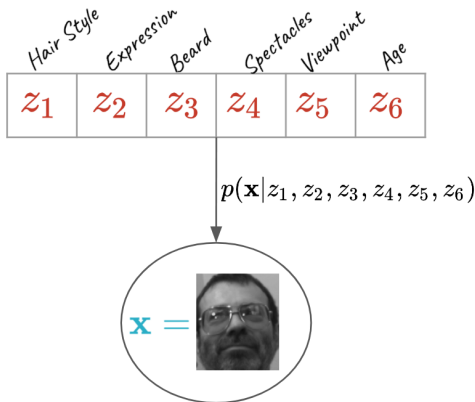
$$p(\mathbf{x}) = \int_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \quad \textbf{Marginalization}$$

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z}) \quad \textbf{Bayes theorem}$$

$$p(\mathbf{x}) = \int_{\mathbf{z}} p(\mathbf{x}|\mathbf{z})p(\mathbf{z}) d\mathbf{z}$$

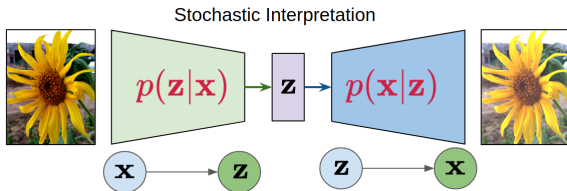
Generative Models

Given a set of points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ from a distribution $p_{\text{data}}(\mathbf{x})$, learn a distribution $p_{\text{model}}(\mathbf{x})$ from the given samples.



Generative Models

Can we use autoencoders directly to model this approach?



$$\mathbf{z} \sim p_z \quad \text{blue } \mathbf{z} \longrightarrow \text{green } \mathbf{x} \quad \mathbf{x} \sim p_x$$
$$p(\mathbf{x}|\mathbf{z}) \quad p(\mathbf{x}) = \int_{\mathbf{z}} p(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Generative Process

$$\begin{aligned}\log p(\mathbf{x}) &= \log \left(\sum_{\mathbf{z} \in \mathcal{Z}} p(\mathbf{x}, \mathbf{z}) \right) \\&= \log \left(\sum_{\mathbf{z} \in \mathcal{Z}} \frac{p(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} q(\mathbf{z}) \right) \\&= \log \left(\sum_{\mathbf{z} \in \mathcal{Z}} \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q(\mathbf{z})} q(\mathbf{z}) \right) \\&= \log \left(\mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} \left[\frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q(\mathbf{z})} \right] \right) \\&\geq \mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} \left[\log \left(\frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q(\mathbf{z})} \right) \right] \\&\geq \mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} [\log p(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} \left[\log \left(\frac{p(\mathbf{z})}{q(\mathbf{z})} \right) \right] \\&\geq \mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} [\log p(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z})), \text{ELBO} \quad \text{Evidence Lower Bound}\end{aligned}$$

Generative Process

$$\begin{aligned}D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) &= \mathbb{E}_{q_z} \left[\log \left(\frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right) \right] \\&= \mathbb{E}_{q_z} \left[\log \left(\frac{q(\mathbf{z})p(\mathbf{x})}{p(\mathbf{z})p(\mathbf{x}|\mathbf{z})} \right) \right] \\&= \mathbb{E}_{q_z} \left[\log \left(\frac{q(\mathbf{z})}{p(\mathbf{z})} \right) \right] - \mathbb{E}_{q_z} [\log p(\mathbf{x}|\mathbf{z})] \\&\quad + \mathbb{E}_{q_z} [\log p(\mathbf{x})]\end{aligned}$$

$$\log p(\mathbf{x}) - D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) = \mathbb{E}_{q_z} [\log p(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}))$$

$$\log p(\mathbf{x}) - D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) \geq \text{ELBO}$$

$$\log p(\mathbf{x}) \stackrel{?}{=} \text{ELBO}$$

$$D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) = 0$$

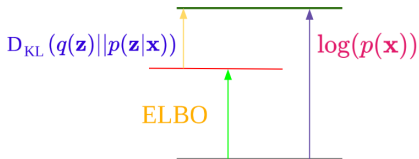
$$q(\mathbf{z}) = p(\mathbf{z}|\mathbf{x}).$$

Variational Auto-encoders

$$\begin{aligned}\log(p(\mathbf{x})) &= \text{ELBO} + D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}|\mathbf{x})) \\ \text{ELBO} &= \mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))] - D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z}))\end{aligned}$$

Maximize ELBO such that $q(z)$ is as close as possible to $p(\mathbf{z}|\mathbf{x})$.
This can be posed as an optimization problem as below.

$$\max \mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))] - D_{\text{KL}}(q(\mathbf{z})||p(\mathbf{z})) \text{ such that } q(\mathbf{z}) = p(\mathbf{z}|\mathbf{x}).$$



Variational Auto-encoders

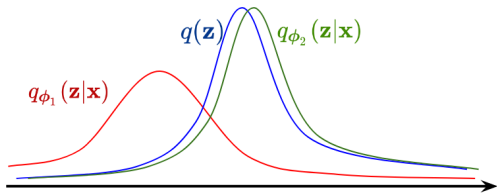
Maximize ELBO such that $q(z)$ is as close as possible to $p(z|\mathbf{x})$.

This can be posed as an optimization problem as below.

$$\max \mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} [\log (p(\mathbf{x}|\mathbf{z}))] - D_{\text{KL}}(q(\mathbf{z}) \| p(\mathbf{z})) \text{ such that } q(\mathbf{z}) = p(\mathbf{z}|\mathbf{x}).$$

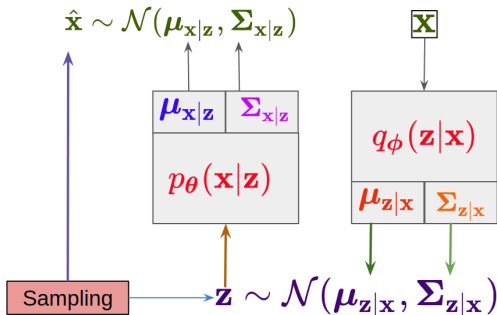
Variational Inference

What if the posterior $p(\mathbf{z}|\mathbf{x})$ is intractable? (This can be seen using Bayes theorem $p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})}$). Suppose $q_{\phi}(\mathbf{z}|\mathbf{x})$ is a tractable probability distribution over the hidden variables $\mathbf{z}$ parameterized by ϕ (variational parameters). Then, pick ϕ so that $q_{\phi}(\mathbf{z}|\mathbf{x})$ is as close as possible to $q(\mathbf{z})$.



Realization of Variational Auto-encoders using NNs

So far we have considered only abstract representations of $p_{\theta}(\mathbf{x}|\mathbf{z})$, $q_{\phi}(\mathbf{z}|\mathbf{x})$, and $q(\mathbf{z})$. Let us assume that $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{z}}, \boldsymbol{\Sigma}_{\mathbf{x}|\mathbf{z}})$, $q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{z}|\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}})$ and $q(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.



Realization of Variational Auto-encoders using NNs

Now, consider the ELBO function $\mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))]$ and use $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{z}}, \sigma_1^2 \mathbf{I})$.

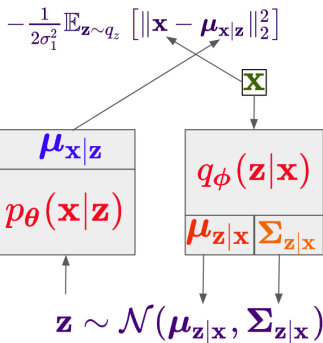
Then, the ELBO function becomes:

$$\mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))] = \mathbb{E}_{\mathbf{z} \sim q_z} \left[\log \left(\mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{z}}, \sigma_1^2 \mathbf{I}) \right) \right]$$

$$\log \left(\mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}|\mathbf{z}}, \sigma_1^2 \mathbf{I}) \right) = -\frac{1}{2\sigma_1^2} \|\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}|\mathbf{z}}\|_2^2 + \text{const.}$$

$$\mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))] = -\frac{1}{2\sigma_1^2} \mathbb{E}_{\mathbf{z} \sim q_z} \left[\|\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}|\mathbf{z}}\|_2^2 \right]$$

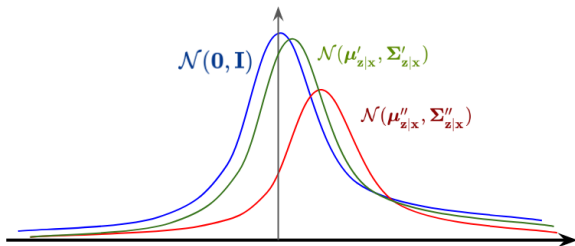
$\Rightarrow$ Reconstruction loss.



Realization of Variational Auto-encoders using NNs

Our goal is to maximize the ELBO function $\mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))]$ such that $q_\phi(\mathbf{z}|\mathbf{x})$ is as close as possible to $q(\mathbf{z})$. That is, we have to minimize $D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||q(\mathbf{z}))$. Now, use $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{z}|\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}})$ and $q(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$.

$$\begin{aligned} D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||q(\mathbf{z})) &= D_{\text{KL}}(\mathcal{N}(\boldsymbol{\mu}_{\mathbf{z}|\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}})||\mathcal{N}(\mathbf{0}, \mathbf{I})) \\ &= \frac{1}{2} \sum_{i=1}^k (\mu_i^2 + \sigma_i^2 - 1 - \log_e(\sigma_i^2)) \end{aligned}$$

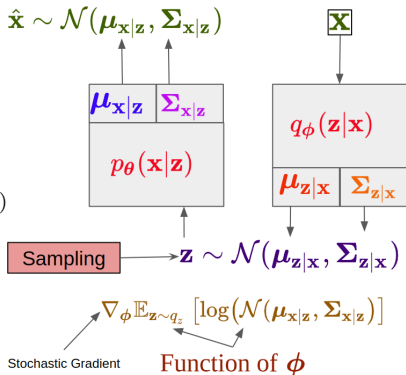


Sampling makes our life difficult

In order to find the optimal weights of the network, we have to find the gradient of the loss function with respect to the parameters ϕ and θ . Let us consider finding the gradient of ELBO function $\mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))]$. We can easily find the gradient of ELBO w.r.t. θ as

$$\nabla_{\theta} \mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))] = \mathbb{E}_{\mathbf{z} \sim q_z} [\nabla_{\theta} \log(p(\mathbf{x}|\mathbf{z}))]$$

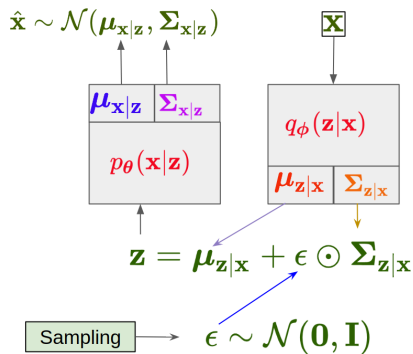
However, here we observe that $\mathbf{z}$ and $\log(p(\mathbf{x}|\mathbf{z}))$ are functions of ϕ . Therefore, finding $\nabla_{\phi} \mathbb{E}_{\mathbf{z} \sim q_z} [\log(p(\mathbf{x}|\mathbf{z}))]$ is not easy as we can not take gradient inside the expectation and this is stochastic gradient.

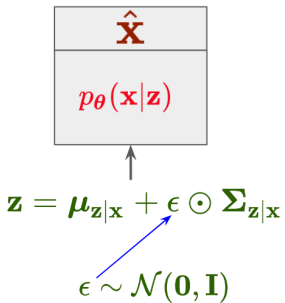


Re-parametrization trick to rescue

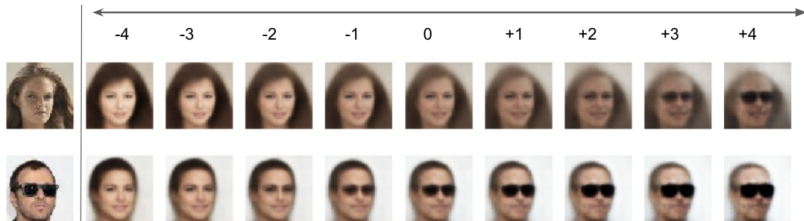
Let ϵ be a random variable such that $\epsilon \sim \mathcal{N}(0, 1)$ and let μ and σ be two constants. Now, let us define a random variable z as $z = \mu + \epsilon\sigma$. Then, find the distribution of the random variable z .

$$\begin{aligned}\mathbb{E}[z] &= \mathbb{E}[\mu + \epsilon\sigma] \\ &= \mathbb{E}[\mu] + \mathbb{E}[\epsilon\sigma] \\ &= \mu + \sigma\mathbb{E}[\epsilon] \\ &= \mu \\ \text{var}(z) &= \text{var}(\mu + \epsilon\sigma) \\ &= \text{var}(\mu) + \text{var}(\epsilon\sigma) \\ &= 0 + \sigma^2\text{var}(\epsilon) \\ &= \sigma^2 \\ z &\sim \mathcal{N}(\mu, \sigma^2).\end{aligned}$$





Kingma and Welling 2014



Murphy 2021

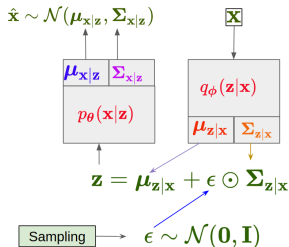
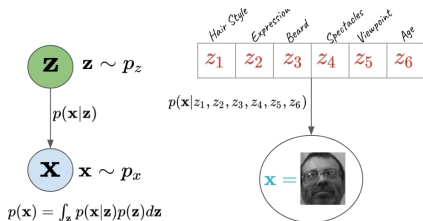
Generative Models

Problem Statement

- ▶ Given a dataset $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ containing samples drawn from an unknown data distribution $p_{\text{data}}(\mathbf{x})$, learn a distribution $p_{\text{model}}(\mathbf{x})$ that is close as possible to the true distribution $p_{\text{data}}(\mathbf{x})$.
- ▶ Draw new samples from the distribution p_{data} by using its approximation p_{model} .

Variational Autoencoders

$$\max \mathbb{E}_{\mathbf{z} \sim q_z} [\log (p_{\theta}(\mathbf{x}|\mathbf{z}))] \text{ such that } q_{\phi}(\mathbf{z}|\mathbf{x}) = q(\mathbf{z}).$$



Problem Statement

- ▶ Given samples $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ drawn from an unknown data distribution $p_{\text{data}}(\mathbf{x})$, learn a distribution $p_{\text{model}}(\mathbf{x})$ that is close as possible to the true distribution $p_{\text{data}}(\mathbf{x})$.
- ▶ Draw new points from p_{data} by using its approximation p_{model} .

Generative Adversarial Networks

- ▶ Our ultimate goal is to draw new samples from the distribution p_{data} .
- ▶ Is it necessary to learn p_{model} that approximate p_{data} ?
- ▶ Can we draw points from p_{model} without explicitly learning p_{model} ?
- ▶ Generative Adversarial Networks [Goodfellow et al. 2014] enables us to draw new samples from p_{data} without explicitly learning p_{model} . Hence,
circumvent maximization of the log-likelihood.